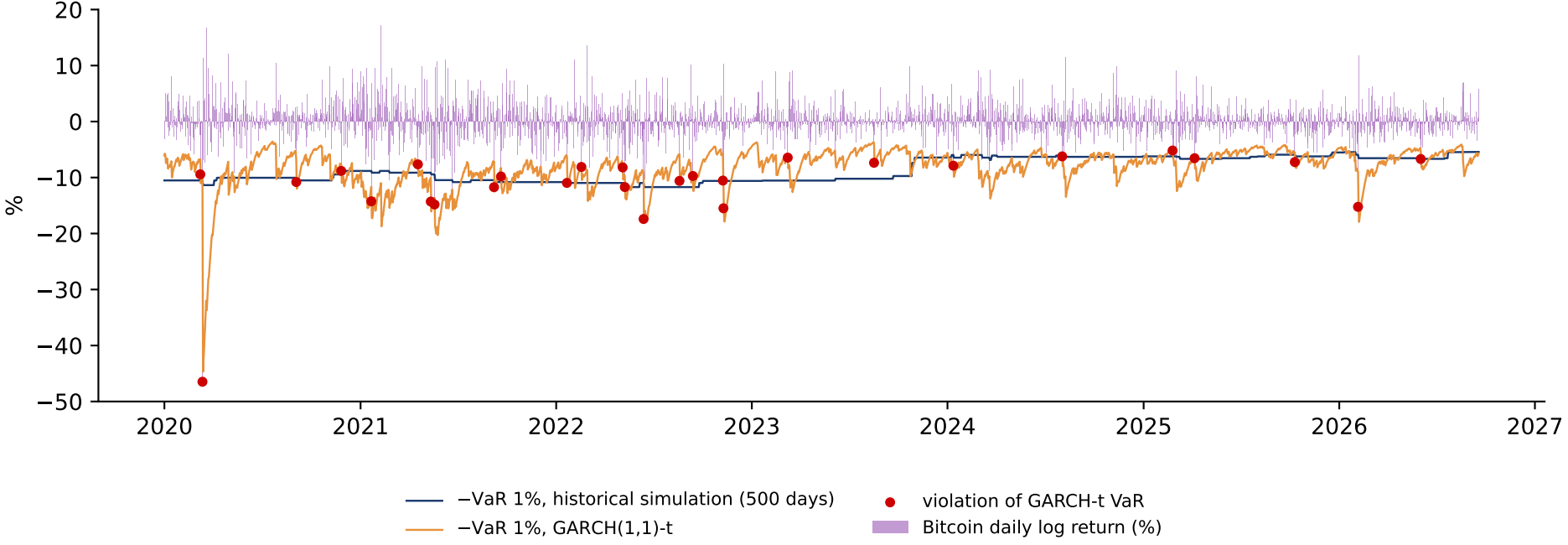

20
10
0
−10
−20
−30
−40
−50
%
2020
2021
2022
2023
2024
2025
2026
2027
−VaR 1%, historical simulation (500 days)
−VaR 1%, GARCH(1,1)-t
violation of GARCH-t VaR
Bitcoin daily log return (%)